



How can such legal complications be mitigated when using generative AI tools for embedded design?

When individuals own intellectual property, such as an operational amplifier or a PLL, they typically file patents or obtain copyrights. However, relying solely on generative AI for IP creation can lead to legal challenges. This occurs because the AI may utilise code from multiple IPs without proper attribution. To mitigate this risk, open frameworks with Creative Commons licences offer legal protection for both commercial and non-commercial use. This becomes increasingly important with the rise of generative AI technologies.

So, what role do SMEs and startups have to play in this ecosystem?

Big time. With this kind of system, we are reducing the manpower requirement in companies to build solutions, thanks to proven silicon designs available for use. This shifts the focus towards system integration skills rather than specific design skills. For startups, this means an easier start as they can leverage existing designs rather than facing difficult challenges from scratch. Hiring individuals with specialised design skills can be challenging due to budget constraints. High tool costs, such as licensing fees, can be prohibitive for startups and SMEs, whose revenues are much lower in comparison. An open AI hardware ecosystem promotes accessibility and openness, encouraging the participation of SMEs, startups, designers, students, and others in the field.

Is this open electronics hardware ecosystem only limited to design?

Not at all. Design, verification, testing, validation, and cross-validation all play a part in this.

When does it come into play in manufacturing?

Manufacturing is another area that

requires discussion. Currently, not many foundries support open source tools and democratisation of chip design is important for the growth of this field. A process design kit (PDK) is needed for creating the manufacturing-ready graphic design system (GDS) file required for tapeout. However, if the foundries themselves do not support it, obtaining a PDK becomes challenging. Foundries need to support FPGAs, and someone has to develop process design tools for designers to use. This conversation needs to happen with many foundries, not just one or two. Currently, only a few foundries support open source designs, making testing possible on those foundries. However, more foundries will likely adapt if there are more industry-ready, silicon-proven designs and interesting innovations emerging. It is a chicken-and-egg problem: if more open source designs become available, foundries will be compelled to provide the necessary support, like PDKs.

If we compare the Indian design ecosystem with the global standard, what do we need to become a design powerhouse?

Certainly, if we observe companies in the technology sector, much of the design work actually occurs in India, indicating a strong talent base in fabless design. However, Indian-owned businesses in this sector are relatively few. Currently, most companies operating in India are either US- or UK-owned, or they are European companies with a presence in India. Indian designers often work for these companies, developing designs that are later sold internationally. So, one change that we need to make is to focus on building Indian-owned companies that can sell their IPs and products globally, not just domestically. The challenge lies in enabling that and making those resources more afford-

able to encourage more people to enter the industry early on. Otherwise, there is a risk of a heavy reliance on Python programming, with fewer advancements in hardware design originating from India.

What role does Digital University Kerala (DUK) play here?

We have implemented a model that completely removes the need for traditional classroom-based teaching and thereby transitions to experiential learning. This means there are no fixed learning hours; it is more about continuous learning, accessible 24/7. Our curriculum is designed around lab-based and project-based learning. This shift is driven by industry demands for targeted skills rather than generic ones. We aim to cultivate problem-solving and creativity without relying on textbooks. Instead, students engage with user manuals, design guides, and reference standards. Additionally, we have established several independent centres of excellence, often registered as companies, to support this ecosystem. These centres include incubators and research facilities focused on commercial product development. Many students participate in these centres while pursuing their masters or PhDs, leading to the creation of startup companies, particularly in the electronic design space. In the last two years alone, we have seen the emergence of around four startup companies, largely driven by students.

In partnership with IEEE, we have launched a programme called Maker CHIPS to democratise the IC design using open source tools. Students and researchers get a chance to learn the art of IC design and manufacture the chips at no cost to the end user. The details of this programme can be found at www.makerchips.org. **EFY**